

FFGFT: More than Two States — How a Qubit Encodes

Johann Pascher

2026

Contents

.1	The Real Question: What Lies Between $ 0\rangle$ and $ 1\rangle$?	2
.2	The Cylindrical Phase Space — FFGFT Geometry of the Qubit	2
.3	Quantum Registers — Exponential Growth through Combination	3
	Product States: Additive Classicality	4
	Entangled States: Multiplicative Growth	4
	Bell States — Maximum Entanglement in Two Qubits	4
.4	Three Ways to More States — a Comparison	6
.5	Quantum Interference — Why Superposition is Useful	6
.6	QND Measurement — Reading Without Destroying	7
	What QND Means	7
	QND as Empirical Proof of Determinism	7
	Limits of QND Measurement	8
	Potential Landscape: Stable Minima, Transitions and Readout Strength	8
.7	Two Levels of the ξ -Hierarchy: Spin Qubit and Photonic Resonator	9
	Spin Qubit: Minimal Resonator at Level $N \approx 8$	9
	Photonic Resonator: Collective Geometry at Level $N \approx 9-10$	9
	Not Restricted to Two States: Infinitely Many Analog Transitions	10
	Comparison of the Two Levels in the ξ -Hierarchy	11
	The Analog Resolution Limit — and Its Continuation in High-Frequency Technology	11
	Above Optics — Three Regimes up to the Pair-Production Threshold	13
	The Cannonball Limit in FFGFT: Scale Incompatibility of Two Torsion Patterns	14
	The Pair-Production Threshold as Natural Endpoint of the ξ -Ladder	14
	Convergence: Carrier Frequency Meets Internal Transition Frequency on ξ -Level $N \approx 8-9$	15
.8	ξ -Level Separation as Stability Principle	16
	The Coupling Formula Between Levels	16
	Fractal Self-Similarity <i>Because of</i> the Damping	16
	Restructuring between Levels: When Does It Happen and When Not?	17
.9	Qudit Register — the Connection to the ξ -Hierarchy	17
.10	The FFGFT Topology Compiler — Geometry of Entanglement	18
.11	Comparison: Classical Bit, Qubit, Qudit, Register	18
.12	Open Connections — Next Steps	19
.13	Summary	19

Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- X-gate angle $\alpha = \pi K_{\text{frak}}$: deviation 0.013 % $\rightarrow 1 - K_{\text{frak}} = 1.33$ % (two places).
- 1 Oct. 2026: CHSH damping $2.8275 \rightarrow 2.82840$, $\Delta\text{CHSH} \sim 10^{-4} \rightarrow 2 \times 10^{-5}$ (three places); thermal margin “four orders of magnitude” $\rightarrow$ factor 40; three or four ξ levels do not reproduce the frequency spans AM-fiber ($2\text{--}3 \times 10^8$) and 1MHz– 10^{15} Hz (10^9); UV (near) 120–40 $\rightarrow$ 413–124 nm; ququart temperature 4 K $\rightarrow T < 1$ K.
- 1 Oct. 2026: π -gate deviation marked as not directly measurable (absorbed by pulse calibration), previously “testable prediction”.

.1 The Real Question: What Lies Between $|0\rangle$ and $|1\rangle$?

Spin-up and spin-down are two points. Two points are unambiguously labelled — one bit of information.

But a **qubit** is not a bit. The real power of the qubit lies not in the fact that it can be $|0\rangle$ or $|1\rangle$ — a classical coin can do that too. It lies in the fact that it can be $|0\rangle$ and $|1\rangle$ *simultaneously* in a superposition — and in a continuous ratio:

$$|\psi\rangle = \cos\frac{\theta}{2} |0\rangle + e^{i\phi} \sin\frac{\theta}{2} |1\rangle. \quad (1)$$

Two parameters: the mixing angle $\theta \in [0, \pi]$ and the relative phase $\phi \in [0, 2\pi)$. Together they parameterise a **sphere surface** — the **Bloch sphere** (geometric representation of all possible single-qubit states as points on a sphere of unit radius).

On this sphere there are *infinitely many points* — not only the two poles $|0\rangle$ (north pole) and $|1\rangle$ (south pole), but the entire equator, all circles of latitude, every point in between. Every point is a valid, physically distinguishable quantum state.

A qubit does not encode 2 states — it encodes *infinitely many*

The two classical poles $|0\rangle$ and $|1\rangle$ are special cases of a continuous family. What distinguishes a qubit from a classical bit is not the number of discrete states — it is the **continuous geometry** of the state space.

The readout interacts with the system — every measurement couples to the torsion configuration and drives it into the nearest stable extreme state ($|0\rangle$ or $|1\rangle$). The result is deterministic — it depends on the exact initial state which we do not fully know before measurement. The apparent “probability” of standard QM describes this epistemic ignorance, not fundamental randomness.

The states themselves are discrete. The continuity lies in the phase space *before* measurement — the readout always delivers one of the two stable poles.

.2 The Cylindrical Phase Space — FFGFT Geometry of the Qubit

The Bloch sphere is the standard representation. In FFGFT (Doc. 034) there is a physically more direct alternative: the **cylindrical phase space**.

A qubit state is described by three geometric coordinates:

$$(z, r, \theta), \quad (2)$$

where:

- $z \in [-1, +1]$: the “height” along the cylinder axis — describes the proportion of the torsion configuration in the direction of $|0\rangle$ vs. $|1\rangle$. ($z = +1$: pure torsion type $|0\rangle$; $z = -1$: pure torsion type $|1\rangle$)
- $r \in [0, 1]$: the radius in the cross-section — indicates the strength of the intermediate torsion, i.e. how far the state is from both poles. ($r = 0$: stable pole; $r = 1$: maximum intermediate torsion)
- $\theta \in [0, 2\pi)$: the azimuthal angle — the **relative phase** between the two basis components.

The relation to the Bloch sphere is direct: $z = \cos \theta_{\text{Bloch}}$ and $r = |\sin \theta_{\text{Bloch}}|$. But the cylinder is not a geometric decoration — it makes gate operations into *geometric transformations* in real space:

Gate	Hilbert space (matrix)	FFGFT cylinder (transformation)
Hadamard H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Swap $z \leftrightarrow r$, rotate phase by $\pi/2$ — moves pole to equator
Phase Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Add π to phase θ — pure rotation about cylinder axis
Bit-flip X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	2D rotation of (z, r) by angle $\alpha = \pi \cdot K_{\text{frak}}$ — fractal damping built in

FFGFT: gates as torsion rotations

In the cylindrical FFGFT formalism every gate operation is a geometric transformation of the torsion vector. The Hadamard gate does not rotate a matrix — it tilts the torsion configuration from the longitudinal axis to the transverse plane. The Z gate rotates the phase θ — it rotates the torsion winding about its own axis.

The factor K_{frak} in the X gate is not a correction factor added afterwards — it follows directly from the fractal spacetime geometry ($D_{\text{frak}} = 2.999867$) and means: every qubit rotation is slightly smaller than π , because spacetime itself minimally deviates from flat 3D geometry. This is a prediction that cannot be measured directly, because it is absorbed by pulse calibration: *(Corrected on 1 Oct. 2026: previously “testable” – a systematic over-rotation of 1.33 % is absorbed by the pulse calibration of real qubits (amplitude adjustment) and cannot be separated from device variation, cf. Doc. 034.)* all π gates should systematically deviate by $\Delta\alpha/\pi \approx 1 - K_{\text{frak}} \approx 1.33\%$ from the ideal value. *(Corrected on 30 Sept. 2026: previously $0.013\% - 1 - K_{\text{frak}} = 100\xi = 1.33\%$; 0.013% is ξ itself.)*

.3 Quantum Registers — Exponential Growth through Combination

A single qubit has infinitely many intermediate states in the torsion phase space, but always delivers one of the two stable poles upon readout — the readout itself is the interaction that drives the state to the next stable extreme. The real power comes through **combination**.

Product States: Additive Classicality

Two independent qubits have together $2 \times 2 = 4$ basis states: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

The general state *without entanglement* (product state) is:

$$|\psi_A\rangle \otimes |\psi_B\rangle = (\alpha_A |0\rangle + \beta_A |1\rangle) \otimes (\alpha_B |0\rangle + \beta_B |1\rangle). \quad (3)$$

This is only two separate Bloch spheres — no gain over two independent qubits. The state space is *additive*: $2 + 2 = 4$ parameters.

Entangled States: Multiplicative Growth

Once the two qubits are *entangled*, the joint state can no longer be written as a product of two individual states. The general two-qubit state requires 4 complex amplitudes (minus normalisation and global phase: $2 \times 4 - 2 = 6$ real parameters):

$$|\Psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle. \quad (4)$$

This is **more** than two independent qubits — entangled qubits carry joint information that is contained in neither individual qubit.

For n qubits the Hilbert space grows exponentially:

$$d(n) = 2^n \quad (5)$$

Real parameters in the general state: $2 \cdot 2^n - 2$.

n qubits	Basis states	Real parameters	Classical equivalent
1	2	2	1 bit
2	4	6	2 bits
3	8	14	3 bits
10	1,024	2,046	10 bits
50	10^{15}	2×10^{15}	50 bits
300	10^{90}	2×10^{90}	$\approx$ number of atoms in the universe

This is the actual quantum advantage: Not a single qubit is powerful — but the collective growth of an *entangled* register.

Bell States — Maximum Entanglement in Two Qubits

The four **Bell states** are the maximally entangled two-qubit states — each a complete orthonormal basis of the entangled subspace:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad (6)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \quad (7)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad (8)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \quad (9)$$

Each of these states has the property: if qubit A is measured, the result of qubit B is immediately determined — regardless of how far away.

In FFGFT (Doc. 023, 035):

$$E^{\text{FFGFT}}(a, b) = -\cos(a - b) \cdot \left(1 - \xi \frac{(\Delta\theta)^2 / \pi^2}{D_{\text{frak}}}\right), \quad (10)$$

the ξ damping reduces the CHSH violation from $2\sqrt{2} \approx 2.82843$ to ≈ 2.82840 — a difference of $\sim 2 \times 10^{-5}$, which explains non-locality locally without eliminating it. (*Corrected on 1 Oct. 2026: previously “to ≈ 2.8275 — a difference of $\sim 10^{-4}$ ” — with the formula above, $\xi = 4/30000$, $D_{\text{frak}} = 3 - \xi$ and the CHSH angles one obtains $S = 2.828404$, hence $\Delta\text{CHSH} \approx 2.4 \times 10^{-5}$; 2.8275 would correspond to $\Delta\text{CHSH} = 9 \times 10^{-4}$.)*

FFGFT: Bell states as joint torsion nodes

In the FFGFT picture a Bell state is not an action at a distance between two qubits — it is a **topologically connected torsion configuration** across two resonators. The two resonators share a common torsion node whose resolution simultaneously determines both ends at every local measurement.

This is not a signal propagating faster than light — it is a common geometric state established at preparation. The local readout interaction at one end deterministically drives the local torsion into a stable pole — and thereby necessarily also determines the state of the other end, because both share the same torsion node. The “global consequence” is not action at a distance, but the geometric coherence of the common node.

The ξ damping (FFGFT correction to Bell correlations) is in this picture the finite extension of the torsion node: the node is not point-like, but has a characteristic length $\sim L_\xi = \xi \cdot \ell_p \cdot (1/\xi)^N$ at the ξ scale level $N \approx 8$.

IBM Quantum hardware validation: Bell states, June 2025

The FFGFT predictions for Bell states were experimentally verified on genuine quantum hardware:

Hardware: IBM Brisbane & IBM Sherbrooke (127 qubits each, Heavy-Hex lattice), access via IBM Quantum Platform / Qiskit Runtime.

Result:

Quantity	Value
Bell state fidelity	97.17 %
FFGFT algorithm vs. QM	Agreement within measurement precision
Repetition variance	0.0001–0.001 (excellent)
Deviation $P(00)$, $P(11)$	1–5 % (hardware noise)

Significance in FFGFT context: The FFGFT amplitudes $E_i(x, t)$ with ξ coupling ($\xi = 1.038 \times 10^{-5}$, Higgs-derived) deliver algorithmically equivalent predictions to standard QM — with improved repeatability (lower variance). The ξ deviation from the ideal Bell correlation ($\Delta\text{CHSH} \approx 2 \times 10^{-5}$ (*Corrected on 1 Oct. 2026: previously $\approx 10^{-5}$ — aligned with the recomputed value 2.4×10^{-5} from the Bell-state section.*)) lies below the current hardware noise threshold and could therefore not yet be directly isolated and measured. This is a precise experimental task for future high-precision Bell tests.

.4 Three Ways to More States — a Comparison

Strategy	More states through	Growth	FFGFT meaning
Superposition (1 qubit)	Continuous superposition of $ 0\rangle$ and $ 1\rangle$	∞ , but 1 bit at measurement	Intermediate torsion on the Bloch sphere
Register (n qubits)	Entanglement of multiple qubits	2^n (exponential)	2^n -dimensional torsion product space; entangled = topologically connected nodes
Qudit ($d > 2$, 1 object)	Higher energy levels, orbital states, exciton number	d (linear) per object, d^n for n qudits	d stable torsion configurations in one resonator

No strategy dominates in all areas

Register entanglement scales most strongly (2^n), but requires n physical objects and error-corrected entanglement operations between them.

Qudit coding gains information *per object* ($\log_2 d > 1$), but the exponential growth is slower.

Superposition is always present — it is the basis for quantum interference, the actual driving force of quantum algorithms.

.5 Quantum Interference — Why Superposition is Useful

Superposition alone would be a mere overlay — the readout would always deliver one of the two stable poles, deterministically dependent on the exact phase state.

What makes superposition into a *resource* is **interference**: amplitudes can constructively (amplify) or destructively (cancel) overlap — depending on phase ϕ .

This is the principle behind all useful quantum algorithms:

1. Prepare the register in a uniform intermediate torsion over all inputs (with the Hadamard gate).
2. Apply a function that increases the phases of the correct answers and decreases those of the wrong ones.
3. Let interference drive the wrong answers into unstable torsion configurations and the correct one into a stable one.
4. The readout drives the system deterministically into the nearest stable pole — which after interference corresponds with overwhelming frequency to the sought answer.

Without interference: Monte Carlo (classical guessing). With interference: Grover ($\sqrt{N}$ instead of N steps), Shor (factorisation in polynomial time).

FFGFT: interference as torsion phase superposition

In FFGFT the phase ϕ is not an abstract complex number — it is the rotation position of the intermediate torsion configuration (Doc. 174).

Constructive interference: two torsion phases coincide — they amplify each other into a geometrically more stable configuration. Destructive interference: two opposite torsion phases cancel each other — the resulting configuration has no stable geometric anchor and is not addressed at the readout interaction.

A quantum algorithm in FFGFT language: a sequence of torsion rotations that aligns the phase configuration of the register so that the sought answer ends in a stable torsion configuration — all others in geometrically unstable ones. The readout is then not a random selection, but a deterministic reaction of the resonator to the readout interaction.

.6 QND Measurement — Reading Without Destroying

What QND Means

A QND measurement reads an observable $\hat{A}$ without changing the corresponding eigenstate. The formal condition:

$$[\hat{A}, \hat{H}_{\text{measurement}}] = 0. \quad (11)$$

The measurement operator $\hat{H}_{\text{measurement}}$ commutes with the measured observable — it does not change the eigenvalue, because it respects the same eigenspace.

Practically this means: the same state can be **repeatedly measured** and delivers the same result each time — because the measurement does not touch the state itself.

The most important example from Doc. 174: **Faraday/Kerr rotation**. A linearly polarised laser beam passes the quantum dot without being absorbed — it rotates its plane of polarisation by $\theta_F \propto S_z$. No photon is absorbed. No energy transfer into the spin mode. The spin state before and after measurement is identical.

QND as Empirical Proof of Determinism

The QND measurement is from the FFGFT perspective far more than a technically useful property — it is an **empirical proof** for the reality of stable torsion configurations.

The argument is direct:

1. Faraday rotation measures S_z without changing the spin state.
2. Repeated measurements always yield the same result.
3. Therefore the state had this value *before every measurement* — definitely, not as an abstract superposition that only “becomes reality” through measurement.
4. The state was real before we looked.

FFGFT: QND measurement and stable torsion configuration

In FFGFT, Faraday rotation is state-preserving because a stable torsion configuration (spin-up = topologically fixed winding in one direction) cannot be driven out of its state through a dispersive, non-resonant coupling.

The photon “sees” the torsion — it rotates its polarisation accordingly — but it transfers no energy into the torsion mode, because it is not resonant. The geometric state is invariant under this coupling.

The philosophical consequence is clear: If a state can be repeatedly measured without disturbance, then it was not “created” by the measurement — it was there before. The standard QM interpretation (“the state exists only through measurement”) is not compatible with QND measurements.

In FFGFT: the torsion configuration is a geometric reality — stable, definite, pre-existent. The measurement reads it off. It does not create it.

Limits of QND Measurement

QND is possible for observables that commute with the measurement operator — typically: S_z (spin component along the quantization axis).

What QND *cannot* do: An intermediate torsion configuration (superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$, i.e. a state that is *not* an S_z eigenstate) is driven by an S_z -QND measurement into an eigenstate — deterministically, depending on the exact initial state, but the intermediate torsion is gone afterwards.

Why this is so is shown by the potential-landscape picture.

Potential Landscape: Stable Minima, Transitions and Readout Strength

FFGFT: torsion potential, transitions and readout strength

The complete picture of qubit states is a **potential landscape with two stable minima** and a saddle between them.

The readout strength decides what happens:

Initial state	Weak coupling	Strong coupling
Stable minimum ($ \uparrow\rangle$ or $ \downarrow\rangle$)	Coupling does not lift state from minimum — QND : state remains, result reproducible	Coupling can lift state from minimum — invasive, state changed
Slope / saddle (superposition)	Even weak coupling is sufficient to drive the state — it “rolls” into the next minimum. Which one : determined by exact phase ϕ (epistemically unknown, deterministic)	Same, just faster — state jumps immediately into next minimum

The decisive insight: There is no principled boundary between “QND-possible” and “QND-impossible” — there are only **more stable and less stable states** in a continuous potential landscape.

A state in the deep minimum is stable against the readout coupling — the minimum must be overcome. A state on the slope is unstable — any coupling, however weak, is sufficient to drive it into the next minimum.

And the “jump” to the final position is not random — it is determined deterministically by the phase of the initial state on the slope: a state near $|\uparrow\rangle$ lands in $|\uparrow\rangle$, a state near $|\downarrow\rangle$ lands in $|\downarrow\rangle$, a state on the saddle ($\theta = \pi/2$) lands depending on the fine phase ϕ .

The apparent “randomness” of standard QM is the epistemic ignorance of this phase — not an ontological randomness of the system.

The picture applies to all d levels of a qudit: not two minima, but d minima — with $d - 1$ saddle points between them.

QND and superposition — no contradiction

A stable eigenstate ($|\uparrow\rangle$ or $|\downarrow\rangle$): sits in the deep minimum — weak coupling reads without driving. QND possible. Result reproducible.

An intermediate torsion ($\alpha|\uparrow\rangle + \beta|\downarrow\rangle$): sits on the slope — any coupling drives into the next minimum. Deterministic, not random. The phase ϕ determines which minimum is reached.

The QND measurement *proves* the reality of stable poles. The disturbing measurement *proves* the instability of intermediate torsions — and the determinism structure of the potential landscape. Together: completely deterministic, geometrically founded picture.

.7 Two Levels of the ξ -Hierarchy: Spin Qubit and Photonic Resonator

Spin Qubit: Minimal Resonator at Level $N \approx 8$

A spin qubit is the absolute minimum of a quantum resonator (Doc. 173): a single electron, a single energy level, two permitted orientations.

The energy scale comes from **quantum confinement** at ξ level $N \approx 8$ (nanometre scale). The spacing between the two spin states in a typical magnetic field ($B \approx 1$ T) lies at:

$$\Delta E_Z = g^* \mu_B B \approx 0.1\text{--}1 \text{ meV}. \quad (12)$$

The thermal energy at room temperature is:

$$k_B T(300 \text{ K}) = 26 \text{ meV}. \quad (13)$$

So $\Delta E_Z \ll k_B T$ — thermal noise constantly knocks the system out of the minimum. Low temperatures ($T < 1$ K, $k_B T < 0.1$ meV) are not optional, but mandatory — not to maintain superposition, but simply to *stay* in one of the two minima.

Photonic Resonator: Collective Geometry at Level $N \approx 9\text{--}10$

A waveguide of lithium niobate (LiNbO_3), a Mach-Zehnder interferometer, a ring resonator — these are resonators made of 10^{18} to 10^{23} atoms in perfect lattice order.

The decisive difference from the spin qubit: **the resonator is not a single particle — it is the macroscopic geometry of the entire crystal.**

The energy scale of an optical photon:

$$E_{\text{photon}} \approx 1\text{--}2 \text{ eV} \quad (780\text{--}1550 \text{ nm}), \quad (14)$$

so:

$$\frac{E_{\text{photon}}}{k_B T(300 \text{ K})} \approx \frac{1 \text{ eV}}{26 \text{ meV}} \approx 40. \quad (15)$$

Thermal noise cannot spontaneously create optical photons — the system sits far below the thermal threshold for optical excitations. Condition 2 from Doc. 173 ($\Delta E \gg k_B T$) is satisfied by a factor of 40.

Why photonics works at room temperature

Not because the photonic system is more robustly built — but because it operates at a higher ξ level of the energy hierarchy.

The **geometric robustness** (Condition 1) is extraordinarily well satisfied by the macroscopic single crystal with $\sim 10^{23}$ lattice sites in perfect order — much more robust than a single quantum dot. Individual phonons (thermal lattice vibrations) barely disturb the collective mode.

The **thermal robustness** (Condition 2) is automatically given by the eV energy scale — thermal photons in the infrared have ~ 25 meV, optical photons have ~ 1 eV: a factor of ≈ 40 in energy, corresponding to a Boltzmann suppression $e^{-40} \approx 4 \times 10^{-18}$.
(Corrected on 1 Oct. 2026: previously “four orders of magnitude safety margin” – $1 \text{ eV}/25 \text{ meV} = 40$, i.e. about 1.6 orders of magnitude in energy.)

Not Restricted to Two States: Infinitely Many Analog Transitions

The spin qubit is reduced to two states — that is its strength (simplicity, control) and its weakness (cooling mandatory, fragile).

The photonic resonator is structurally different: It is *not* restricted to the minimal-two-state structure of spin-up/down.

In an MZI (Mach-Zehnder interferometer) the photon runs through two paths simultaneously — with a continuously adjustable phase ϕ between the paths. The phase is an analog quantity that can continuously traverse $[0, 2\pi)$. The interference result is $\cos^2(\phi/2)$ — a continuous transfer function, not a discrete two-level structure.

Analog or digital — a choice of operating mode

A photonic resonator can operate in **both modes**:

Digital: Phase set to $\phi = 0$ or $\phi = \pi$ — then transmission is 1 or 0. Two discrete states, like a classical bit.

Analog: Phase continuously between 0 and 2π — then transmission is a continuous number between 0 and 1. Infinitely many intermediate states, all stable at room temperature, because the energy scale is thermally unreachable.

Analog readout of these transitions requires *no* cooling — but it requires geometric precision of the resonator (Condition 1 from Document 173).

This is the fundamental difference from the spin qubit: With the spin qubit the analog information (the phase ϕ on the slope of the potential landscape) lies in a thermally unstable intermediate state — T_2 collapses at room temperature within nanoseconds.

With the photonic resonator the analog information lies in the phase of the photon — and photons are **thermally stable**, because their energy scale is far above $k_B T$. The state on the “slope” here is not a fragile minimum — it is a geometrically stabilized mode parameter of the crystal.

Comparison of the Two Levels in the ξ -Hierarchy

	Spin qubit	Photonic resonator
ξ level	$N \approx 8$ (nm scale)	$N \approx 9\text{--}10$ (μm –mm scale)
Resonator	1 electron in nm cage	$10^{18}\text{--}10^{23}$ atoms as collective crystal
Energy scale	$\Delta E \approx 0.1\text{--}1\text{ meV}$	$E_{\text{photon}} \approx 1\text{--}2\text{ eV}$
$\Delta E/k_B T$ (300 K)	$\approx 0.004\text{--}0.04$ (too small!)	≈ 40 (robust)
Cooling needed?	Yes, $T < 1\text{ K}$ mandatory	No, room temperature
Analogue states	Fragile: $T_2 \sim \text{ns}$ at 300 K	Stable: phase thermally unreachable
Discreteness from	Quantum confinement (1 particle)	Macroscopic mode geometry
Typical application	Digital qubit, quantum gate	Analogue signal processing, photonic computing, 6G

FFGFT: two levels of the ξ hierarchy — same physics, different scales

Spin qubit and photonic resonator are in FFGFT not fundamentally different systems — they are resonators at different levels of the ξ scaling ladder.

At level $N \approx 8$: the confinement energy minimum lies at meV — not thermally robust at 300 K. At level $N \approx 9\text{--}10$: the collective mode minimum lies at eV — completely thermally robust.

The ξ parameter generates a universal energy hierarchy: each level carries the energy by a factor of $1/\xi \approx 7500$ higher than the previous. From meV (level 8) to eV (level 9–10): the factor is $\sim 10^3\text{--}10^4$ — consistent with ξ scaling.

This explains not only photonics at room temperature — it explains why biological systems use photonic mechanisms (photosynthesis, bird navigation via light) and not spin qubit systems for quantum effects: nature works at the thermally robust level.

The Analog Resolution Limit — and Its Continuation in High-Frequency Technology

The photonic resonator is thermally robust — but only as long as one does not resolve too finely. As soon as one wants to distinguish analog intermediate values, the energy space E_{photon} is divided into N distinguishable levels. The energy difference between adjacent levels:

$$\Delta E_{\text{level}} = \frac{E_{\text{photon}}}{N}. \quad (16)$$

Condition 2 from Document 173 applies inexorably:

$$\Delta E_{\text{level}} \gg k_B T \implies N \ll \frac{E_{\text{photon}}}{k_B T}. \quad (17)$$

At room temperature and optical photons ($E \approx 1\text{ eV}$):

$$N_{\text{max}}(300\text{ K}) \approx \frac{1\text{ eV}}{26\text{ meV}} \approx 40. \quad (18)$$

The maximum analog resolution at room temperature is about 40 distinguishable levels — corresponding to $\log_2(40) \approx 5.3$ bits per photon (Shannon limit for this energy level).

The photonic system does not escape the thermal problem — it shifts it to a finer scale. More resolution requires either cooling (which gives the same argument as for the spin qubit) or an even higher energy scale.

The entire history of high-frequency technology as an ascent in the ξ -hierarchy

This is not a new insight — it is the oldest engineering wisdom of communications technology, now geometrically grounded.

High-frequency technology has been climbing the same ξ -hierarchy for a hundred years. At each frequency level the signal-to-noise ratio improves — not because engineers built better circuits, but because $\Delta E/k_B T$ grows with each level.

Technology	Frequency	$E = hf$	$E/k_B T$	N -level	Thermal regime
AM broadcast-ing	1 MHz	4×10^{-9} eV	1.5×10^{-7}	≈ 5.5	Thermally fully dominated; noise $\gg$ signal
VHF/FM	100 MHz	4×10^{-7} eV	1.5×10^{-5}	≈ 6.4	Still strongly noise-limited; high-power transmitters required
Microwave / Radar	10 GHz	4×10^{-5} eV	1.5×10^{-3}	≈ 7.3	Noise manageable; low-noise amplifiers required
5G / mmWave	30 GHz	1.2×10^{-4} eV	5×10^{-3}	≈ 7.5	Transition regime; cooling of most sensitive stages (radio telescopes)
THz	1 THz	4×10^{-3} eV	0.15	≈ 7.8	Difficult zone: $E \sim k_B T$; that is why THz technology is so demanding
Near-infrared / LiDAR	200 THz	0.8 eV	31	≈ 8.8	Thermally robust; room temperature without difficulty
Optics / fiber	300 THz	1.2 eV	46	≈ 9.0	Thermally fully stable; $N_{\max} \approx 46$ analog levels
Short-wave UV	1,000 THz	4 eV	154	≈ 9.4	High analog resolution; $N_{\max} \approx 150$ levels

The unifying relation:

Each level in the ξ -hierarchy carries the energy scale a factor $1/\xi \approx 7500$ higher than the previous one. From $N = 6$ (AM, MHz) to $N = 9$ (optics, THz) three levels bridge the factor $(1/\xi)^3 \approx 4 \times 10^{11}$ — the frequency of optical fiber, however, lies only $\sim 2\text{--}3 \times 10^8$ times above AM broadcasting; the three ξ levels and the actual span disagree by a factor of $\sim 1500\text{--}2000$. (Corrected on 1 Oct. 2026: previously “and indeed the frequency of optical fiber lies $\sim 10^8$ times above AM broadcasting” – $(1/\xi)^3 = 4.2 \times 10^{11}$, whereas $(2\text{--}3) \times 10^{14} \text{ Hz} / 10^6 \text{ Hz} = (2\text{--}3) \times 10^8$; the N column of the table does not follow the factor 7500 per level either: $1 \text{ MHz} \rightarrow 100 \text{ MHz}$ gives $\Delta N = 0.9$ there, with 7500 per level it would be $\ln 100 / \ln 7500 = 0.52$.)

In this picture optical fiber is not a new technology — it is the consistent continuation of the same trend to the next ξ -level.

And the resolution limit applies at every level: whoever wants to resolve more than ~ 40 analog levels at optical wavelengths must cool — just as the radio engineer in 1930 shielded his receiver and optimized his amplifier in order to find a signal in thermal noise.

The problem is the same. The scale shifts. The physics does not change.

In FFGFT: The universal resolution limit per temperature level:

$$N_{\max}(T, N_{\xi}) \approx \frac{E_0 \cdot (1/\xi)^{N_{\xi}-N_0}}{k_B T} \quad (19)$$

with $E_0 = k_B T_0$ at the reference level N_0 . Each ξ -level higher multiplies the attainable resolution by the factor $1/\xi \approx 7500$.

Above Optics — Three Regimes up to the Pair-Production Threshold

Optics ($N \approx 9$) is not the end of the ξ -ladder — but it is the end of the usable zone for guided photonic information transmission. What follows above it traces a clear sequence:

Range	Energy	N -level	λ	What happens to the resonator
UV (near)	3–10 eV	≈ 9.3	413–124 nm	Photoionization sets in; electrons are knocked out of outer shells. The resonator is damaged by its own signal.
Soft X-rays	100 eV – 1 keV	≈ 10	12–1 nm	λ approaches the lattice spacing ($d \approx 0.3$ nm). Bragg scattering at individual lattice planes — no longer a homogeneous medium. Waveguiding collapses.
Hard X-rays	1–100 keV	≈ 10.5	1–0.01 nm	$\lambda < d_{\text{lattice}}$: the photon flies through gaps in the lattice. Matter partially transparent. No refractive index, no resonator.
Gamma / nuclear transitions	>100 keV	≈ 11	<0.01 nm	Interaction only with atomic nuclei. Entirely different ξ -level — nuclear levels instead of electronic levels.
Pair-production threshold	>1 MeV	≈ 11.3	—	Pair production, natural upper limit.

(Corrected on 1 Oct. 2026: row UV (near), previously $\lambda = 120\text{--}40$ nm — $\lambda = 1239.84 \text{ eV nm}/E$ gives 413–124 nm for 3–10 eV; 120–40 nm would correspond to 10–31 eV.)

The Cannonball Limit in FFGFT: Scale Incompatibility of Two Torsion Patterns

In the particle picture the argument is clear: the photon becomes a cannonball flying through gaps in the lattice as soon as $\lambda \lesssim d_{\text{lattice}}$.

In FFGFT there are no cannonballs and no waves as separate concepts — there are only **torsion configurations of the field on different scales**.

FFGFT: Waveguiding as Scale Resonance of Two Torsion Patterns

The crystal is a **periodic torsion structure**: a spatially repeated pattern of stable field nodes on the ξ -level $N \approx 7\text{--}8$ (atomic scale, $d \approx 0.3\text{ nm}$).

The photon is a **propagating torsion wave**: a coherent field configuration with characteristic length scale λ .

Waveguiding works precisely when these two scales are compatible — when the photon's torsion wave perceives the periodic lattice structure as a *collective* homogeneous torsion environment. This requires $\lambda \gg d_{\text{lattice}}$.

Three regimes in FFGFT language:

Condition	Particle picture	FFGFT picture
$\lambda \gg d_{\text{lattice}}$	Wave sees homogeneous medium; guided propagation	Torsion wave couples collectively to lattice nodes — effective refractive index emerges as average of node density
$\lambda \approx d_{\text{lattice}}$	Bragg scattering at individual lattice planes	Torsion wave begins to resolve individual lattice nodes — collective coupling breaks into single-node coupling; useful for crystallography, destructive for waveguiding
$\lambda \ll d_{\text{lattice}}$	Cannonball flies through lattice gaps; matter transparent	Scale incompatibility: torsion wave is too fine to "see" the lattice nodes — no coupling, no guiding

The crucial point: The particle picture and the FFGFT wave picture are *the same geometric argument* — formulated at different description levels. FFGFT shows why: because there is no fundamental difference between particle and wave. Both are torsion configurations. The difference is only the **scale of the configuration relative to the scale of the environment**.

The "wave-particle duality" of standard QM is not a duality in FFGFT — it is a single geometry that appears as wave or particle depending on the observation scale.

The Pair-Production Threshold as Natural Endpoint of the ξ -Ladder

The pair-production threshold at $E > 2m_e c^2 = 1.022\text{ MeV}$ is not a technical failure — it is a **geometric limit of the ξ -hierarchy** for carrier systems.

In FFGFT: the photon is a propagating torsion configuration with energy E_{Photon} . An electron and a positron are two **stable elementary torsion nodes** — the most fundamental stable configurations on the scale of elementary particles.

Their rest energy is $m_e c^2 = 0.511$ MeV each. Together: $2m_e c^2 = 1.022$ MeV.

As soon as $E_{\text{Photon}} \geq 2m_e c^2$: the torsion energy of the photon exceeds the threshold to create two new stable configurations. The photon *reorganizes* into two more stable local torsion nodes — not because it is destroyed, but because that is the energetically possible geometric transition.

Pair-production threshold in the ξ -hierarchy

The threshold $E = 2m_e c^2 \approx 1$ MeV corresponds to the ξ -level $N \approx 11.3$ — the scale at which elementary particles exist as stable torsion configurations.

This is the natural upper limit for photonic information transmission: above this threshold the photon is no longer a pure carrier — it becomes a particle generator. The usable zone of the ξ -hierarchy for guided information transmission lies between:

$$N \approx 5.5 \text{ (AM broadcasting, MHz)} \quad \text{to} \quad N \approx 9.3 \text{ (near UV, eV)}. \quad (20)$$

The span from 1 MHz (AM) to 10^{15} Hz (UV) amounts to $10^9 \approx (1/\xi)^{2.3}$ in frequency; four ξ levels, $(1/\xi)^4 \approx 3 \times 10^{15}$, exceed it by a factor of $\sim 3 \times 10^6$. *(Corrected on 1 Oct. 2026: previously "Four ξ -levels, factor $(1/\xi)^4 \approx 3 \times 10^{15}$ in frequency — precisely the span from 1 MHz (AM) to 10^{15} Hz (UV)" — $10^{15}/10^6 = 10^9$, $(1/\xi)^4 = 3.2 \times 10^{15}$.)*

FFGFT: Unified View of Wave, Particle and Lattice Interaction

In FFGFT three apparently different phenomena express the same geometric logic:
Refractive index: Collective coupling of a torsion wave to many lattice nodes simultaneously — emergent phenomenon when $\lambda \gg d_{\text{lattice}}$.

Bragg scattering / cannonball limit: Transition from collective to single-node coupling when $\lambda \approx d_{\text{lattice}}$ — the same scale incompatibility from particle and wave perspectives.

Pair production: Torsion energy exceeds the threshold for new stable configurations — the photon reorganizes into two torsion nodes (electron + positron).

All three are transitions between torsion configurations on different ξ -levels. The ξ -hierarchy is not only a length-scale partition — it is a partition of the **stable coupling regimes** between torsion configurations.

Convergence: Carrier Frequency Meets Internal Transition Frequency on ξ -Level $N \approx 8-9$

So far two frequencies have been carried in parallel, without making their relation explicit:

Carrier frequency: the external frequency of the light that transports information — the frequency by which we climb the ξ -ladder.

Transition frequency: the internal resonance frequency of a quantum resonator — the frequency corresponding to an energy jump between two levels: $\nu_{\text{internal}} = \Delta E/h$.

These two frequencies are fundamentally different. But on ξ -level $N \approx 8-9$ something remarkable happens: they **converge**.

The internal transition frequency of a quantum dot on ξ -level $N \approx 8$ (nanometer scale) lies at $\Delta E \approx 0.5-2$ eV — precisely in the optical range, 1550 nm to 700 nm wavelength.

Fiber-optic communication operates at 1550 nm ($E \approx 0.8$ eV) — exactly in the window where the internal transitions of the InGaAs semiconductor structures lie that serve as lasers, amplifiers and detectors.

Convergence on ξ -level $N \approx 8-9$

Fiber-optic communication at 1550 nm works so naturally because on ξ -level $N \approx 8-9$ two scales coincide:

	Scale	Energy
Carrier frequency (fiber)	$\lambda = 1550$ nm	0.8 eV
Internal transitions (InGaAs QD)	$R \approx 3-5$ nm	0.7–1.2 eV
Exciton Bohr radius (CdSe)	$a_X \approx 5.5$ nm	~ 1 eV

This is not a materials-science convention — it is geometrically forced by the ξ -hierarchy. The same scale level that determines the refractive index of the waveguide also determines the internal energy levels of the active components. Carrier and resonator speak the same language because they live on the same ξ -level. And this is why waveguiding breaks down above the optical window not only from lattice geometry — but also because the carrier frequency leaves the ξ -level on which the active material components operate.

.8 ξ -Level Separation as Stability Principle

The Coupling Formula Between Levels

Between different ξ levels, the coupling strength decreases exponentially with the level difference ΔN :

$$\Gamma_{\Delta N} \propto \xi^{\Delta N}. \quad (21)$$

For $\Delta N = 1$: $\xi^1 \approx 1.333 \times 10^{-4}$ (weak but non-zero). For $\Delta N = 2$: $\xi^2 \approx 1.78 \times 10^{-8}$ (extremely weak). For $\Delta N = 4$: $\xi^4 \approx 3.2 \times 10^{-16}$ (practically non-existent).

Fractal Self-Similarity *Because of* the Damping

The fractal hierarchy of nature is not stable *despite* ξ damping — it is stable *because of* it.

Were ξ not small, all levels would strongly couple to each other. Every change at one level would immediately restructure all other levels. There would be no stable atoms, no stable molecules, no stable cells — only a single chaotic coupled system of all scales simultaneously.

Because $\xi \ll 1$, the levels can exist **quasi-independently**: each level has its own stable internal dynamics, barely disturbed by the other levels.

FFGFT: ξ damping as origin of the natural hierarchy

The existence of separate, stable structural levels in nature — quarks, nucleons, atoms, molecules, cells, organisms, planets, galaxies — is not a random property of matter.

It follows directly from ξ level separation: each level is practically invisible to the next-but-one. The coupling falls exponentially with level distance. This is the mechanism that makes the fractal self-similarity of nature *stable*.

Restructuring between Levels: When Does It Happen and When Not?

The question “can one level restructure another?” now has a precise answer:

Yes, when:

- ΔN is small (≤ 1) — the energy scales are close enough that the coupling $\xi^{\Delta N}$ can be compensated by resonant enhancement.
- Or the energy on level N_1 exceeds the *threshold* for a new stable configuration on level N_2 — as in pair production ($\gamma \rightarrow e^+ + e^-$).

No, when:

- $\Delta N \geq 2$ — the coupling $\xi^{\Delta N} \leq 10^{-8}$ is too weak to cause stable restructuring.
- The energy is not sufficient to overcome the threshold of the target configuration.

The Natural Hierarchy as a Geometric Stability System

Atoms are not restructured by phonons ($\Delta N \approx 1$, coupling $\sim \xi$: weak, but not zero — which is why decoherence exists, but atoms still remain stable).

Cells are not restructured by atomic vibrations ($\Delta N \approx 4$, coupling $\sim \xi^4 \approx 10^{-16}$: absolutely negligible).

Galaxies are not restructured by phonons ($\Delta N \approx 8$, coupling $\sim \xi^8 \approx 10^{-32}$: non-existent).

The hierarchy of nature — from quarks to galaxies — is in FFGFT no accident of matter distribution. It is the direct consequence of ξ -level separation: each level is geometrically protected against disturbances from far-away levels. Self-similarity and stability are two sides of the same ξ -geometry.

.9 Qudit Register — the Connection to the ξ -Hierarchy

From Doc. 174: a quantum dot of dimension d encodes $\log_2 d$ bits. What happens when one entangles n such qudits?

The total state space has dimension d^n . For $d = 4$ (spin-orbital ququart) and n qudits:

$$d^n = 4^n = (2^2)^n = 2^{2n}. \quad (22)$$

This is equivalent to $2n$ qubits in a register of size n — double the information density per object.

Qudit density at ξ level $N \approx 8$

A quantum dot with $R = 3$ nm at $T < 1$ K has at least 4 thermally stable states (spin-orbital ququart, $d = 4$). (*Corrected on 1 Oct. 2026: previously “at 4 K” — $k_B T(4\text{ K}) = 0.34\text{ meV}$, while the Zeeman splitting at 1 T is only 0.1–1 meV ($g = 2$: 0.12 meV); the criterion $\Delta E \gg k_B T$ is not met for the spin half of the ququart at 4 K, at 4 K only with a much larger B field.*) A register of $n = 10$ such qudits has dimension $4^{10} = 1,048,576$ — the same as 20 classical qubits, but only 10 physical objects at ξ level $N \approx 8$.

.10 The FFGFT Topology Compiler — Geometry of Entanglement

A practical problem with entangled registers: entanglement gates between non-adjacent qubits are expensive — they require many SWAP operations (which accumulate errors and cost time).

FFGFT (Doc. 034) offers a geometric solution: the **FFGFT topology compiler**.

Standard compilers minimise the number of SWAPs. The FFGFT compiler instead minimises the cumulative **fractal path length** of all entanglement operations. Since the fractal damping term $\exp(-\xi \cdot \ell / D_{\text{frak}})$ is distance-dependent (larger physical distance ℓ means stronger ξ damping of coherence), critically interacting qubits must be *physically placed closer* to each other.

This leads to a new chip architecture: not laid out according to logical circuit optimisation, but according to **geometric coherence maximisation**.

Connection to the adaptive hybrid architecture

The adaptive hybrid architecture (Doc. 173 conclusion) selects resonators according to task: quantum dots for local synaptic operations, Ising machines for global optimisation.

The FFGFT topology compiler is the linking element: it decides which qubits/qudits must physically sit close together (strong entanglement) and which may remain further apart (weakly, tolerantly damped entanglement).

The learning AI on this hardware does not only learn *what* is to be computed — it also learns *which geometric arrangement* executes the computation most coherently. Not through programmed topology specifications, but through geometrically informed learning in the FFGFT compiler.

.11 Comparison: Classical Bit, Qubit, Qudit, Register

	Class. bit	Qubit	Qudit (d)	n qubits
State space	2 points	Bloch sphere (∞)	$\mathbb{CP}^{d-1}$ (∞)	$\mathbb{CP}^{2^n-1}$ (∞)
Readout result	1 bit (deterministic)	1 bit (deterministic, depending on phase state)	$\log_2 d$ bits (deterministic)	n bits
Parallel processing	none	Interference over 2 paths	Interference over d paths	Interference over 2^n paths
Resource for advantage	—	Phase ϕ	$d - 1$ phases	Entanglement + phases
FFGFT picture	stable/unstable torsion	Torsion on Bloch sphere	d stable torsion levels	Torsion network, topologically connected

.12 Open Connections — Next Steps

Open questions for further chapters

1. Fractal gate corrections measurable?

The FFGFT compiler says: all π pulses deviate by $\Delta\alpha/\pi \approx K_{\text{frak}} - 1 \approx -1.33\%$ (*Corrected on 30 Sept. 2026: previously $0.013\% - 1 - K_{\text{frak}} = 100\xi = 1.33\%$; 0.013% is ξ itself.*) from the ideal π . Current superconducting qubit systems achieve gate fidelity $> 99.9\%$. That is just at the boundary to see this systematic deviation. Is there a systematic bias in the π pulse length in current calibration data?

2. Bell states and ξ scale level:

The ξ damping in Bell correlations ($\Delta\text{CHSH} \approx 2 \times 10^{-5}$) depends on $N \approx 8$. (*Corrected on 1 Oct. 2026: previously $\Delta\text{CHSH} \approx 10^{-4}$ – the damping formula of the Bell-state section gives $S = 2.828404$, hence $\Delta\text{CHSH} \approx 2.4 \times 10^{-5}$.*) For Bell tests with *macroscopic* objects (atoms, molecules) the scale level N would increase — and with it the damping. Is there an N -dependent suppression of Bell violations for different system sizes?

3. Qudit entanglement and torus topology:

Two entangled qudits ($d = 4$) correspond in FFGFT to two topologically connected spin-orbital torsion configurations. What is the geometric description of this connection in the toroidal field? Is there a conserved quantity (topological invariant) that quantifies the entanglement depth?

4. Quantum error correction and ξ granularity:

The surface code needs ~ 1000 physical qubits per logical qubit. In a qudit-based register with $d = 4$ one needs only half as many physical objects for the same logical information. How does the error correction threshold change in the FFGFT framework when the natural noise floor $\xi \approx 1.333 \times 10^{-4}$ applies as the lower error bound?

.13 Summary

Conclusion: More than Two States

A single qubit has infinitely many intermediate states in the torsion phase space — but the states themselves are **discrete**: the readout always delivers one of the two stable poles, deterministically determined by the exact phase state at the moment of readout. The apparent “probability” of standard QM is epistemic ignorance of this state — not fundamental randomness of the system.

The readout interacts with the system — but not always destructively. Faraday/Kerr rotation (Doc. 174) is a QND measurement: it reads the state without changing it, because the coupling commutes with the spin operator. That the same state can be repeatedly measured is the empirical proof that it *existed before* measurement — definitively and really. The result remains deterministic in every case.

The actual quantum advantage comes through three routes:

1. Superposition: Continuous intermediate torsion with phase ϕ enables interference — the geometric foundation of all useful quantum algorithms. Interference aligns the

register so that the sought answer ends in a stable torsion configuration; the readout follows that deterministically.

2. Register entanglement: n qubits span a 2^n -dimensional torsion space. Entanglement means: the torsion nodes of the qubits are topologically connected — reading one node necessarily also determines the others. Bell correlations are locally explained by ξ damping without eliminating non-locality.

3. Qudit coding: A resonator with $d > 2$ stable torsion levels delivers $\log_2 d > 1$ bits per object at readout. Spin-orbital ququarts ($d = 4$) at ξ level $N \approx 8$ double the information density of a register.

In FFGFT all three strategies are expressions of the same fundamental principle: stable torsion configurations in geometrically precise resonators — discrete from inside, analogously controllable from outside, deterministic at readout.

Bibliography

- [1] Kouwenhoven, L. P. et al. *Excitation spectra of circular, few-electron quantum dots*. Science **278**, 1788–1792 (1997).
- [2] Loss, D. & DiVincenzo, D. P. *Quantum computation with quantum dots*. Phys. Rev. A **57**, 120–126 (1998).
- [3] Veldhorst, M. et al. *An addressable quantum dot qubit with fault-tolerant control-fidelity*. Nature Nanotechnol. **9**, 981–985 (2014).
- [4] Muhonen, J. T. et al. *Storing quantum information for 30 seconds in a nanoelectronic device*. Nature Nanotechnol. **9**, 986–991 (2014).
- [5] Atatüre, M. et al. *Observation of Faraday rotation from a single confined spin*. Nature Phys. **3**, 101–105 (2007).
- [6] Grangier, P., Levenson, J. A. & Poizat, J. P. *Quantum non-demolition measurements in optics*. Nature **396**, 537–542 (1998).
- [7] Wang, C. et al. *Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages*. Nature **562**, 101–104 (2018).
- [8] Reck, M. et al. *Experimental realization of any discrete unitary operator*. Phys. Rev. Lett. **73**, 58–61 (1994).
- [9] Shannon, C. E. *A mathematical theory of communication*. Bell Syst. Tech. J. **27**, 379–423 (1948).
- [10] Knill, E., Laflamme, R. & Milburn, G. J. *A scheme for efficient quantum computation with linear optics*. Nature **409**, 46–52 (2001).

Addendum R61: Status of ξ_{Higgs} (7 September 2026)

The quantity $\xi_{\text{Higgs}} \approx 1.038 \times 10^{-5}$ used in this document is **withdrawn** (R61, Doc. 190-Archive-2). Justification and canonical replacement ($\xi_0 = 4/30000$): see addendum in Doc. 174.